1 Label-free graph representation

Input graph
$G = (V, E, X)$

Node features X

Adjacency A

	v_1	v_2	⋯	v_n
v_1	0	1	⋯	0
v_2	1	0	⋯	1
⋯	⋮	⋮	⋱	⋮
v_n	0	1	⋯	0

Two augmented views

edge dropping + feature masking

independent augmentation

Shared GRACE encoder
g_ψ

× labels not used

Embedding H^1

Embedding H^2

Frozen normalized representation $\bar{H}$

after training: $\bar{H}$ = RowNorm[$g_\psi(X,A)$]

2 Label-agnostic granular-ball partition

Geometry-only construction in frozen embedding space $\bar{H}$

Initialization
$B_0 = V$

Recursive bi-partition

Further recursive splitting

Embedding space $\bar{H}$ covered by granular balls $\mathcal{B} = \{B_j\}_{j=1}^{K}$

selected ball B_j

fixed budget: $K \le B_{max} = 64$

Balls partition all N nodes; only labeled training nodes vote. Construction is label-agnostic; circle overlaps are projection artifacts.

Node counts and labeled proportions are schematic and not to scale.

3 Leave-one-out ball-level label consensus

(a) Select a granular ball B_j

v

(b) Leave-one-out consensus

$p_{j,-v}$ (exclude v)

v's own vote removed

$$q_{j,-v} = (1-\beta)p_{j,-v} + \beta e_{m_{j,-v}}$$

If the majority ties: $q_{j,-v} = p_{j,-v}$
(tie-safe; no arbitrary one-hot class)

(c) Construct a soft target t_v

noisy one-hot + local consensus = soft target t_v

$$t_v = (1 - \alpha_{j,-v})e_{\tilde{y}_v} + \alpha_{j,-v}q_{j,-v}$$

$\alpha_{j,-v}$ measures vote concentration
$\beta = 0.5$ in the fixed configuration

4 X-only MLP training and inference

Training with graph-derived targets
for each training node $v \in V_L$

2-layer MLP fθ

node feature x_v

precomputed t_v

Soft cross-entropy prediction vs. t_v (Eq. 13)

Inference after training (transductive)

trained MLP

node feature x

class probabilities
no graph access

Class 1 · Class 2 · Class 3 · Class 4 · Class 5 · non-voting node · excluded self-vote · data flow · contrastive alignment

X-only inference: no A, no $\bar{H}$, no neighbors